INDUSTRIAL INTERNSHIP REPORT

Online Internship Program in Embedded and IoT – 2026

Project Report on

IoT-Based Smart Temperature and Humidity Monitoring System

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Internship Platform: upSkill Campus / The IoT Academy

Internship Year: 2026

Details to complete before submission: Add the college logo, college name, guide name and actual internship dates. This report uses a proposed project topic; confirm it matches the project approved by your internship provider.

EXECUTIVE SUMMARY

This report describes a proposed educational prototype titled **IoT-Based Smart Temperature and Humidity Monitoring System**. The system demonstrates how an embedded controller, an environmental sensor and wireless communication can be combined to collect and view measurements. The design uses an ESP32 development board and a DHT11 temperature and relative-humidity sensor.

The DHT11 senses the surrounding temperature and humidity. The ESP32 reads the sensor output, checks that the reading is valid and sends the information through Wi-Fi to an IoT dashboard or cloud platform. The measurements can then be viewed remotely using a connected computer or mobile device. An optional LCD can display the values locally.

The project connects the four-week internship learning sequence with a practical application. Week 1 introduces IoT architecture and applications. Week 2 covers IoT devices, platforms and the development of embedded systems. Week 3 introduces deployment and challenges. Week 4 focuses on embedded-system circuit design. These concepts are applied in the proposed monitoring system.

The report includes the problem statement, existing and proposed solutions, system architecture, circuit connections, program flow, test plan, learning outcomes and future improvements. The system is intended for learning and demonstration and is not presented as a certified industrial measurement device.

Project status and honesty note: The project topic and design are proposed for this report. No physical prototype readings, test screenshots or cloud dashboard evidence have been provided. The performance chapter therefore contains test cases and a result-recording template. Replace it with actual observations after building and testing the system.

Keywords: Internet of Things, Embedded Systems, ESP32, DHT11, Wi-Fi, sensor monitoring, dashboard.

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The page sequence follows the supplied 16-page USC/UCT sample report. The project-specific material is written for the selected Embedded and IoT topic.

1. PREFACE

Industrial internship training is an important part of engineering education because it connects classroom concepts with practical applications. It gives students an opportunity to understand how technical knowledge is used to identify problems, design solutions and evaluate results. It also helps students improve communication, documentation, teamwork and time-management skills.

I participated in the Online Internship Program in Embedded and IoT – 2026 through upSkill Campus and The IoT Academy. The course learning resources were organised into weekly sections. The topics included IoT introduction, architecture and applications; IoT devices and platforms; the rise of embedded systems; IoT deployment and challenges; and embedded-system circuit design.

The proposed project selected for this report is an IoT-Based Smart Temperature and Humidity Monitoring System. It is designed to demonstrate how a sensor can collect environmental information, how a microcontroller can process the data and how a wireless connection can make the readings available on a dashboard.

The internship helped me understand the relationship between hardware and software in an IoT solution. It also encouraged me to explore sensor interfacing, microcontroller programming, network connectivity and systematic testing. The report records the learning and proposed design in a structured format.

I express my gratitude to my college, faculty members, internship coordinators, upSkill Campus and The IoT Academy for providing learning resources and guidance. I also thank everyone who supported my learning process. I would encourage juniors to practise each concept, maintain weekly notes, test circuits carefully and document their work honestly.

Note: Add the names of your faculty guide and internship mentor, if required by your college.

2. INTRODUCTION

The Internet of Things (IoT) is a network of physical objects that contain sensors, processing capability and communication interfaces. These objects collect information from their surroundings and exchange it with other devices or software platforms through a network. IoT systems are used in homes, agriculture, healthcare, transportation, industries and smart-city applications.

An IoT solution commonly contains four functional parts: sensing, processing, communication and application. Sensors measure physical quantities such as temperature, humidity, pressure, light or motion. A microcontroller or embedded computer reads the sensor values and performs basic processing. A communication interface transfers the information to a local server or cloud platform. The application presents the information through dashboards, charts, alerts or reports.

Embedded systems are specialised computing systems designed to perform a particular task within a larger product. They combine hardware and software and are often designed to operate with limited memory, processing power and energy. Microcontrollers such as the ESP32 are widely used in learning and prototyping because they combine processing, input/output pins and wireless connectivity in a compact board.

IoT and embedded systems are closely related. The embedded controller interacts with the physical world, while IoT communication makes the collected information accessible to remote users or services. In the proposed project, the DHT11 sensor measures temperature and humidity, the ESP32 reads the sensor and Wi-Fi provides the communication link.

The main purpose of this report is to explain the proposed system using the terminology and structure of the supplied internship report template. The following sections describe the internship context, problem, solution, design, testing approach and future scope.

2.1 ABOUT UNICOVERGE TECHNOLOGIES PVT LTD (UCT)

The supplied sample report describes UniConverge Technologies Pvt Ltd as an organisation established in 2013 and working in digital transformation and industrial solutions. It identifies technology areas including IoT, cyber security, cloud computing, machine learning, communication technologies such as 4G/5G/LoRaWAN, Java full stack, Python and front-end development. Refer to the official internship material for the latest company description.

2.1 ABOUT UNICOVERGE TECHNOLOGIES PVT LTD (CONTINUED)

The sample report introduces the UCT IoT platform, UCT Insight. It describes the platform as supporting the development of IoT applications and providing information for business or process monitoring. The sample states that the platform uses Java for its backend and ReactJS for its front end, with support for MySQL and NoSQL databases.

The sample report lists connectivity through industry-standard protocols such as MQTT, CoAP, HTTP, Modbus TCP and OPC UA. It also describes cloud and on-premises deployment options. The listed platform capabilities include dashboard creation, analytics and reporting, alerts and notifications, integration with third-party applications such as Power BI, SAP and ERP systems, and a rule engine.

The sample report also describes a Smart Factory Platform called Factory Watch. It presents this platform as supporting production and asset monitoring, overall equipment effectiveness (OEE), predictive maintenance and the analysis of machine-generated data. It notes a modular architecture and a software-as-a-service (SaaS) model.

These examples show how IoT data can support operational visibility. A sensor reading by itself is only a measurement. When it is collected over time, stored, displayed and analysed, it can help users identify changes and make decisions. The proposed temperature and humidity monitor demonstrates this basic data path on a smaller educational scale.

The company and platform descriptions on this page are summarised from the supplied USC/UCT sample internship report. They are included to match the sample format and should be checked against current official material before external publication.

2.1 INDUSTRIAL IoT APPLICATION AREAS

The supplied sample report mentions LoRaWAN-based solutions in areas such as agriculture technology, smart cities, industrial monitoring, smart street lighting and smart water, gas and electricity metering. LoRaWAN is a low-power wide-area networking technology that can be used when devices need to send small amounts of data over long distances while conserving battery power.

The sample also describes predictive maintenance solutions that use embedded systems, Industrial IoT and machine-learning technologies to monitor machine health and estimate remaining useful life. Predictive maintenance aims to identify signs of equipment degradation before an unexpected failure occurs. Sensor information such as vibration, temperature, current and operating hours can contribute to machine-health analysis.

Examples of IoT applications include:

- Smart agriculture: monitoring soil moisture, temperature and environmental conditions.
- Smart homes: controlling lighting, appliances and security devices.
- Healthcare: monitoring selected patient or environmental parameters using appropriate certified devices.
- Industrial monitoring: collecting machine status and process measurements.
- Smart utilities: reading water, gas or electricity consumption remotely.
- Environmental monitoring: observing temperature, humidity, air quality and weather conditions.

The proposed project is a small environmental-monitoring example. It does not replace calibrated industrial instruments. Its purpose is to help a student understand how sensing, embedded processing and network communication work together.

2.2 ABOUT UPSKILL CAMPUS (USC) AND THE IoT ACADEMY

The supplied sample report states that upSkill Campus, The IoT Academy and UniConverge Technologies Pvt Ltd collaborated to facilitate the internship process. It describes upSkill Campus as a career-development platform intended to support learners through upskilling, projects, interaction with industry experts and career-growth services.

The sample report describes The IoT Academy as the EdTech division of UCT and mentions long executive certification programmes in collaboration with EICT Academies at IIT Kanpur, IIT Roorkee and IIT Guwahati. This information is reproduced as a summary of the supplied sample document; refer to the provider's current website for updated details.

The internship course screenshots provided by the student show the following learning sequence:

- Week 1: IoT introduction, architecture and applications; The IoT Academy learning resources; weekly progress report.
- Week 2: IoT devices, IoT devices and platforms, applications of IoT, and the rise of embedded systems and IoT as emerging technologies; IoT Test 1; weekly report.
- Week 3: IoT deployment and challenges; Embedded and IoT e-book; weekly report.
- Week 4: Embedded Systems Training – Circuit Design Parts 1, 2 and 3; IoT Test 2; weekly report.
- Final submission: GitHub live-session guidance, sample internship report, IoT Test 3, final project submission and feedback.

The final submission screenshot instructs students to create a GitHub repository named **upskillcampus**, upload the project code and report, use suitable file names, and update the report with working GitHub links. The Java file name shown in the screenshot is an example for a Java project; an IoT project should use its actual source-code file names.

2.3 OBJECTIVES, 2.4 REFERENCES AND 2.5 GLOSSARY

Objectives of the internship

- Understand the basic concepts, architecture and applications of the Internet of Things.
- Identify common IoT devices, sensors, controllers and platforms.
- Understand the role of embedded systems in IoT applications.
- Learn how sensors interface with a microcontroller.
- Understand wireless connectivity and the flow of data to an application.
- Practise basic circuit design, programming, testing and technical documentation.
- Improve problem-solving, communication and career-readiness skills.

References

- USC/UCT, *Sample Industrial Internship Report*, supplied internship template (16 pages).
- upSkill Campus / The IoT Academy, *Free Summer Internship in Embedded and IoT – 2026*, course materials and screenshots supplied by the student.
- Espressif Systems, ESP32 documentation and technical reference: https://docs.espressif.com/
- Adafruit, DHT sensor guide and library documentation: https://learn.adafruit.com/

Glossary

Term	Meaning
IoT	Internet of Things
ESP32	Microcontroller development platform with Wi-Fi and Bluetooth capability
DHT11	Digital temperature and relative-humidity sensor
GPIO	General Purpose Input/Output
Wi-Fi	Wireless local-area networking technology
MQTT	Message Queuing Telemetry Transport
RH	Relative Humidity
API	Application Programming Interface
Dashboard	Visual interface for displaying measurements

3. PROBLEM STATEMENT

Temperature and humidity are important environmental parameters in places such as classrooms, laboratories, storage rooms, greenhouses and small equipment rooms. If these conditions are not observed regularly, an increase or decrease may go unnoticed until it affects comfort, stored materials or equipment operation.

A basic manual method requires a person to visit the location and read a thermometer or hygrometer. This approach can be inconvenient when measurements are required frequently or when the location is difficult to access. A standalone display provides local information but does not automatically make the readings available to a remote user.

The problem addressed by the proposed project is how to measure temperature and relative humidity using a low-cost embedded system and make the readings available through a wireless connection. The system should be simple to assemble, easy to understand and suitable for demonstrating the basic stages of an IoT application.

The proposed solution uses a DHT11 sensor connected to an ESP32 microcontroller. The ESP32 reads the sensor output and transmits the data through Wi-Fi to an IoT dashboard or cloud service. The dashboard can display the latest values and, where the platform supports it, store historical readings for viewing as a trend.

Expected project outcomes

- Acquire temperature and relative-humidity measurements at a defined interval.
- Read and process sensor data using an embedded controller.
- Transmit valid readings through Wi-Fi to an IoT platform.
- Display measurements in a clear dashboard or optional local LCD.
- Identify basic limitations such as sensor accuracy, Wi-Fi availability and power requirements.

Scope limitation: The DHT11 is a low-cost educational sensor. The proposed system is intended for learning and demonstration, not safety-critical control, medical use or certified industrial monitoring.

4. EXISTING AND PROPOSED SOLUTION

4.1 Existing approaches

Common approaches include manual measurement using a thermometer and hygrometer, standalone digital temperature-humidity meters, and network-connected environmental monitoring devices. Manual measurement is simple but depends on a person taking readings. A standalone meter makes local readings convenient but may not provide remote access or automatic data history. Commercial IoT monitoring systems can provide remote access and alerts, but may be more expensive or complex for a small learning project.

4.2 Limitations of existing approaches

- Manual readings may be infrequent and may not provide continuous observation.
- A local-only display cannot be viewed remotely unless a person is physically present.
- Commercial monitoring equipment may require additional cost, configuration or subscription services.
- Different sensors and platforms have different accuracy, connectivity and data-retention capabilities.

4.3 Proposed solution

The proposed design uses an ESP32 board and DHT11 sensor. The sensor measures temperature and relative humidity. The ESP32 reads the digital output, checks the returned values and uses its Wi-Fi capability to send the data to an IoT dashboard. An optional LCD can be added for local display. The design uses readily available educational components and can be extended with alerts or data logging.

4.4 Value addition

- Remote viewing of sensor readings through an internet-connected dashboard.
- Low-cost prototype suitable for learning embedded programming and IoT data flow.
- Expandable design that can accept additional sensors or alert functions.
- Practical demonstration of the relationship between sensing, processing and communication.

4.5 Code and report submission

GitHub repository name required by the course instruction: **upskillcampus**. Suggested project source file: **Smart_Temperature_Humidity_Monitor.ino**. Suggested report file: **Smart_Temperature_Humidity_Monitor_Ranjitha_PL_USC_UCT.pdf**. Replace these suggestions if your provider gives a different naming rule. Add the actual repository and file URLs after uploading; do not use placeholder links as final links.

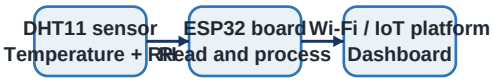
5. PROPOSED DESIGN / MODEL

The proposed system is organised into sensing, processing, communication and presentation stages. The DHT11 sensor measures the surrounding temperature and humidity. The ESP32 acts as the processing and communication unit. It reads the sensor, checks the measurement and connects to a Wi-Fi network. An IoT platform receives the values and presents them to the user.

Main hardware and software elements

- ESP32 development board: reads sensor data and provides Wi-Fi connectivity.
- DHT11 sensor: measures temperature and relative humidity and sends digital data.
- USB cable / regulated power source: supplies power to the controller.
- Wi-Fi router or hotspot: provides network access for data transfer.
- IoT dashboard or cloud platform: receives and displays the measurements.
- Optional LCD: displays current readings locally.

Design flow



Sensor measurements are processed by the ESP32 and transmitted wirelessly.

Figure 1: High-level architecture of the proposed IoT monitoring system.

The design separates the sensor from the network application. This makes it possible to change the dashboard platform without changing the basic sensing principle. The exact platform, account configuration and data protocol should be selected during implementation. For example, an IoT service may use MQTT or HTTP, depending on its supported interface.

5.1 HIGH-LEVEL DIAGRAM, 5.2 LOW-LEVEL DIAGRAM AND 5.3 INTERFACES

Low-level circuit connection



DHT11 VCC → ESP32 3V3
DHT11 DATA → ESP32 GPIO 4
DHT11 GND → ESP32 GND
ESP32 powered by USB / regulated supply

Verify the pin labels on the exact sensor module before connecting.

Figure 2: Example connection arrangement for a DHT11 module and ESP32.

The example uses GPIO 4 as the sensor data pin. Connect the DHT11 VCC to the ESP32 3.3 V pin, DATA to GPIO 4 and GND to GND. Some DHT11 modules contain a built-in pull-up resistor; a bare sensor may require an external pull-up resistor according to its datasheet. Always verify the exact module pin labels and voltage requirements before powering the circuit.

Interfaces and data flow

- Sensor interface: digital data connection between DHT11 and ESP32.
- Network interface: Wi-Fi connection from ESP32 to the local access point.
- Application interface: MQTT or HTTP endpoint supported by the selected IoT platform.
- User interface: dashboard showing temperature, humidity and, if supported, historical charts.

The sensor sampling interval must follow the sensor documentation. The ESP32 should handle Wi-Fi disconnection and invalid sensor readings without displaying misleading values. Network credentials should be stored securely and must not be published in a public source-code repository.

5.3 WORKING PRINCIPLE AND PROGRAM LOGIC

Working sequence

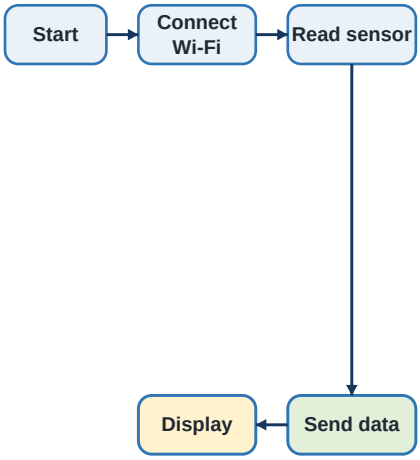


Figure 3: Simplified program flow for the monitoring system.

The program starts by configuring the serial monitor, sensor interface and Wi-Fi connection. After the network is available, the ESP32 requests a reading from the DHT11. The program checks whether the reading is valid. If it is valid, the values are formatted and sent to the configured IoT service. The cycle repeats after a suitable delay. If the sensor returns an error or the network is unavailable, the program should report the condition and retry safely.

Illustrative pseudocode

```
START
Initialise serial communication and DHT11 sensor
Connect ESP32 to Wi-Fi
WHILE system is running:
    Read temperature and humidity
    IF reading is valid:
        Display values locally / on serial monitor
        Send values to IoT platform
    ELSE:
        Record sensor error and retry later
    Wait for the next sampling interval
END
```

Implementation note: This is pseudocode, not a tested source program. The actual code depends on the selected dashboard service, its authentication method and its library/API. Add the final working code and screenshots to the GitHub repository after testing.

6. PERFORMANCE TEST

Performance testing checks whether the proposed system meets its intended requirements. For this project, the main constraints are sensor reading validity, measurement stability, Wi-Fi availability, successful data transfer, display clarity and power stability. Because no actual hardware test results were supplied, the following table is a test plan to be completed during implementation.

6.1 Test plan / test cases

Test ID	Test case	Expected outcome
T1	Power on ESP32	Board starts and runs the program.
T2	Read DHT11 sensor	Valid temperature and humidity values are obtained.
T3	Disconnect sensor	Program identifies an invalid reading or sensor error.
T4	Connect to Wi-Fi	ESP32 connects to the configured network.
T5	Send data to platform	Dashboard receives and displays the submitted values.
T6	Disconnect Wi-Fi	Program reports connection loss and retries safely.
T7	Continuous operation	Repeated readings occur without unexpected reset.

6.2 Test procedure

- Assemble the circuit with the power disconnected and verify all wiring.
- Connect the ESP32 to a computer and upload the program using the selected development environment.
- Open the serial monitor and check whether the sensor readings are valid.
- Configure the Wi-Fi credentials and the selected IoT platform endpoint.
- Observe whether readings appear on the dashboard and record timestamps and values.
- Repeat the tests after simulating sensor and network interruptions.

6.3 Performance outcome

Record actual results in the following format: date/time, test ID, observed value or message, pass/fail status, and corrective action. Do not enter estimated or invented readings. Accuracy should be checked against a suitable reference instrument if accuracy is part of the project requirement. The DHT11 has limited accuracy and response characteristics compared with higher-grade sensors.

7. MY LEARNINGS

The internship learning sequence helped me build an understanding of the Internet of Things and embedded systems. I learned that an IoT solution is not only a sensor connected to the internet; it is a complete system that includes sensing, embedded processing, communication, data handling and a user-facing application.

The study of IoT architecture helped me understand the roles of devices, gateways, networks, cloud platforms and dashboards. Learning about IoT devices and platforms introduced the importance of selecting suitable sensors, controllers and communication methods. The topics on deployment and challenges highlighted practical issues such as connectivity, security, power consumption, reliability and maintenance.

The embedded-system circuit-design topics helped me understand that hardware connections must be planned carefully. Correct voltage levels, pin selection, grounding and sensor wiring are essential for reliable operation. The proposed project brings these concepts together through a sensor, ESP32 controller and Wi-Fi-based data display.

These skills can support my future learning in electronics and communication engineering, embedded development, IoT applications and system testing. I also gained practice in organising technical information, preparing a report and planning a GitHub submission.

8. FUTURE WORK SCOPE

- Use a more accurate sensor, such as a suitable DHT22 or calibrated industrial sensor, if the application requires better measurement performance.
- Add an LCD or OLED screen for local display.
- Add alerts when temperature or humidity crosses user-defined limits.
- Store readings in a database and show daily, weekly and monthly trends.
- Add battery operation and power-saving modes for portable monitoring.
- Improve security by using protected credentials, encrypted communication and access control.
- Extend the system with additional sensors such as light, air-quality or soil-moisture sensors.
- Validate the prototype against a reference instrument and document real test results.

Conclusion: The proposed IoT-Based Smart Temperature and Humidity Monitoring System is a suitable learning model for understanding the basic connection between embedded systems and IoT. It demonstrates the planned flow from environmental sensing to wireless data presentation. The prototype should be built, tested and documented with actual evidence before final internship submission.

Before submitting: Add your college logo and name, guide details, actual implementation photographs, working source code, real test observations, and the final GitHub repository/file links. Confirm that this project title is accepted by your internship provider.